

■ ■ Track Autopsy

AI-Powered Satirical Sonic Pathology Blotter & Forensic Audio Engine

GitHub: <https://github.com/Anuj-9009/track-autopsy>

Tech Stack

TypeScript	React 19	Vite 8.3	Tailwind CSS v4	Google Gemini 3.6 Flash	Spotify Web API
HTML5 Canvas Oscilloscope	REST / JSON Schema				

Type	Full-Stack Forensic Web Application & Autonomous Musicological Investigator
Language	TypeScript 88% • HTML/CSS 10% • Shell 2%
AI & Data	Google Gemini 3.6 Flash (Structured Output Schema) • Spotify Web API
GitHub	github.com/Anuj-9009/track-autopsy

Description

Track Autopsy is a production-grade satirical forensic laboratory and archival document web application that conducts clinical post-mortem examinations on culturally flatlined pop anthems, one-hit wonders, and algorithmic casualties. Users enter any song title and artist into a vintage police blotter docket, triggering a synchronized multi-stage musicological inquest. The application queries Google Gemini 3.6 Flash using a strict structured JSON schema to extract real Billboard Hot 100 historical peak longevity data, original recording studio origins, typewriter transcription of the lethal earworm hook lyric, and a 4-band acoustic spectrum breakdown. Simultaneously, the Spotify Web API is queried via a client credentials flow to fetch high-resolution album artwork and release cataloging. Built with a zero-failure architecture, an internal mathematical generator procedurally produces 100% complete autopsy dossiers offline if API keys are omitted.

Key Features

- **Autonomous Musicological Inquest:** Deep forensic inquiries via Gemini 3.6 Flash extracting verified Billboard chart peaks, weeks at #1, and cultural flatline timelines.
- **Live Spotify Web API Integration:** Seamless client credentials token caching and dynamic album artwork retrieval.
- **Interactive CRT Oscilloscope:** Real-time dual audio frequency visualization on HTML5 Canvas with play/pause and 0.5x, 1.0x, 2.0x sweep speed controls.
- **4-Band Forensic Frequency Breakdown:** Isolated acoustic spectrum mapping real instruments (Sub-Bass 20–60Hz, Low-Mids 200–500Hz, Presence 2–4kHz, High Air 10–20kHz) and true LUFS dynamic squashing metrics.
- **Chain of Custody & Evidence Locker:** 4-tier evidentiary transfer ledger (2-inch master tape, radio promo CD-R, stream transcode, warped vinyl) and Section 9 Custodian Oath with mechanical red ink stamp.
- **Coroner Protocols (Directive 84-M):** Clinical codification of 5 statutory death thresholds and a live searchable ICD-Audio Diagnostic Directory with active case pinning.
- **Zero-Failure Offline Resiliency:** Dual-engine design pairing live AI models with an internal deterministic mathematical generator for 100% offline uptime.
- **Bespoke Academic Credentials:** Satirical doctorates calculated dynamically for the examined artist (e.g., Dr. Gotye, D.A.P. — Doctor of Acoustic Pathology).

Architecture & Design

Modern single-page application architecture built with React 19, TypeScript, and Vite, styled using an authentic mid-century police blotter and coroner docket design system (Space Mono typography, aged manila paper palette #ece8df, distressed borders, and red ink rubber stamps). The data layer features an asynchronous pipeline that concurrently dispatches musicological research queries to Gemini 3.6 Flash and asset requests to the Spotify Search API. The AI engine enforces strict `responseSchema` typing (`Type.OBJECT`) to guarantee predictable, type-safe JSON serialization without hallucinations. A client-side `localStorage` credentials bridge allows users to supply their own API keys via an in-app Config modal, keeping sensitive secrets strictly client-side without committing `.env.local` to version control.

Technical Challenges Solved

- **Reliable Structured LLM Extraction:** Formulated comprehensive JSON schemas and system prompts ensuring Gemini 3.6 Flash consistently outputs verified Billboard metrics, real instruments, and humorous forensic verdicts without JSON parsing failures.
 - **Zero-Crash Offline Resiliency:** Built a deterministic hash-based procedural generator (`utils.ts`) that guarantees complete, rich forensic dossiers and frequency spectra even with missing API keys or zero internet connectivity.
 - **High-Performance Canvas Waveform Engine:** Engineered a 60-FPS dual-phase sine wave oscilloscope on HTML5 Canvas with adjustable temporal sweep speed multipliers and realistic cathode-ray phosphor glow.
 - **Full Narrative Cohesion Across 4 Navigation Tabs:** Unified the data schema so that the exact studio name, hook lyric, frequency bands, and disposal routing orders persist seamlessly between the Case Registry, Waveform Dissection, Chain of Custody, and Coroner Protocols.
 - **Print-Ready Archival CSS Architecture:** Built dedicated `@media print` stylesheets stripping UI chrome and formatting the digital dossier into a sterile, black-and-white physical medical examiner autopsy report.
 - **Strict Credential Isolation:** Architected local environment management ensuring API secrets remain untracked in git, accompanied by a clean `.env.example` template for open-source contributors.
-

Project Verification & Performance

The project was validated for production deployment with zero TypeScript compilation warnings and instantaneous client-side hot module replacement. Bundle optimization via Vite achieves sub-second initial page loads with code splitting and gzipped CSS. Fully responsive layout verified across desktop 12-column asymmetric grids and mobile tab bars.
